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Evaluating Deep Learning-Based Intrusion Detection for Adversarial Attacks on IoT Medical Devices in Cloud-Integrated Hospital Networks

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Table of Contents

Contents

Table of Contents 2
Introduction 3
2. Aims & Objectives / Problem Context 4
AIM: 4
OBJECTIVES: 4
PROBLEM CONTEXT : 5
3. Literature Review 5
SECTION 1 IoT Healthcare Security Landscape 5
SECTION 2 Deep Learning in Cybersecurity Applications 5
SECTION 3 Adversarial Machine Learning Attacks 6
SECTION 4 Cloud-Integrated Healthcare Security 7
SECTION 5 Knowledge Gaps & Research Opportunities 7
4. **Methodology** 8
Research Approach 8
Data Collection Methods 8
Tools and Technologies 8
Experimental Design 8
Limitations and Challenges 9
5. SEPLi (Social, Ethical, Professional, and Legal Issues) 9

SOCIAL IMPACT 9
ETHICAL CONSIDERATIONS 9
PROFESSIONAL STANDARDS 9
LEGAL COMPLIANCE 10
6. Project Plan 10
Project Overview 10
Interactive Gantt Chart 10
Project Structure and Timeline 11
Phase 1: Literature Review & Theoretical Framework (Days 1-4) 11
Phase 2: Ethics Approval & Data Acquisition (Days 3-6) 11
Phase 3: Methodology Development & Tool Setup (Days 5-8) 12
Phase 4: Baseline System Implementation (Days 7-10) 12
Phase 5: Adversarial Attack Development (Days 9-14) 12
Phase 6: Deep Learning Model Development (Days 12-17) 12
Phase 7: Experimental Evaluation & Analysis (Days 16-21) 12
Phase 8: Thesis Writing & Submission (Days 20-25) 13
Critical Milestones Timeline 13
Resource Allocation Strategy 13
Risk Management Framework 14
Conclusion 16
Restate Significance: 16
Summarize Contributions: 16
Future Implications: 16
Final Impact Statement: 16
References 16

Introduction

Healthcare digitalization has revolutionized patient care delivery yet simultaneously exposed critical national infrastructures to unprecedented cybersecurity threats. February 2024 Change Healthcare ransomware attack exemplifies this vulnerability, disrupting 15 billion annual healthcare transactions and affecting one in every three patient records nationwide. This incident caused $6.3 billion in claims processing losses within three weeks and compromised 100 million

Americans' protected health information, marking the largest healthcare data breach in U.S. history.

IoT medical devices integrated with cloud infrastructures present particularly acute security gaps, as 95% of significant healthcare breaches in 2023 originated from third-party systems. The 600% increase in healthcare data breaches since 2020 demonstrates escalating adversarial targeting of medical infrastructure. Deep neural networks, while effective for intrusion detection with 97.28% accuracy rates, remain vulnerable to adversarial perturbations that can evade detection through imperceptible input modifications.

This research addresses the critical gap between adversarial machine learning threats and deep learning-based security solutions in healthcare IoT environments. Given that hospitals require sustained operations without core technology systems for extended periods, developing robust adversarial-resistant intrusion detection systems becomes paramount for patient safety and healthcare system continuity.

2. Aims & Objectives / Problem Context

AIM:

To comprehensively evaluate and enhance deep learning-based intrusion detection systems' resilience against sophisticated adversarial attacks targeting Internet of Medical Things devices in cloud-integrated healthcare environments.

OBJECTIVES:

Analyze adversarial attack vectors and vulnerability patterns across IoMT device architectures in cloud-integrated hospital networks through systematic literature synthesis.

Develop novel deep learning models integrating CNN LSTM architectures with attention mechanisms to detect sophisticated attack signatures while maintaining computational efficiency.

Evaluate model robustness against adversarial perturbations using benchmark datasets

CICIoMT2024, WUSTL EHMS 2020) with comprehensive performance metrics including accuracy, false positive rates, and detection capabilities.

Validate proposed frameworks through comparative analysis against existing intrusion detection systems, demonstrating superior performance in real-time threat detection scenarios.

Optimize federated learning approaches for distributed IoMT security, ensuring privacy preservation while maintaining detection accuracy across heterogeneous healthcare device networks.

PROBLEM CONTEXT :

Healthcare organizations report 99% containing known exploited vulnerabilities in IoMT devices, with ransomware attacks causing $22 million damage and affecting 53% of critical systems. Traditional intrusion detection systems fail against adversarial attacks exploiting 6.2% of IoMT devices, compromising patient safety, data integrity, and regulatory compliance in cloud integrated hospital environments.

3. Literature Review

SECTION 1 IoT Healthcare Security Landscape

The Internet of Medical Things IoMT) has revolutionised healthcare delivery while simultaneously introducing unprecedented cybersecurity vulnerabilities that threaten patient safety and data integrity. Recent investigations reveal alarming statistics: 92% of healthcare organisations experienced at least one cyberattack in 2024, with 75% of incidents caused by hacking activities Doménech et al., 2025; Shaikh et al., 2025 . The vulnerability landscape is particularly concerning, with studies identifying an average of 6.2 vulnerabilities per medical device and 53% of connected medical devices containing critical security flaws Mirkhail & Xinyou, 2025 .

Critical analysis of current security frameworks reveals fundamental inadequacies in addressing IoMT-specific threats. Traditional network security approaches fail to accommodate the unique characteristics of medical devices, including resource constraints and real-time operational requirements Shaikh et al., 2025 . The 2024 Change Healthcare ransomware attack exemplified these vulnerabilities, where attackers leveraged lateral movement through interconnected medical systems to compromise over 100 critical applications Shaikh et al.,

2025 .

Cybersecurity incident trends demonstrate exponential growth, with healthcare ransomware attacks surging by 278% between 2018 2023, while data breach costs reached an average of $10.93 million per incident Madathil et al., 2025 . Current frameworks inadequately address the heterogeneous nature of IoMT environments, where devices from different manufacturers operate with varying security protocols, creating systemic vulnerabilities that enable sophisticated multi-vector attacks.

SECTION 2 Deep Learning in Cybersecurity Applications

Deep learning architectures have emerged as promising solutions for IoMT intrusion detection, with hybrid models demonstrating superior performance over traditional approaches. CNN architectures excel at spatial feature extraction from network traffic patterns, achieving 97.89% accuracy on standardized datasets Zhang et al., 2025 . However, critical analysis reveals that standalone CNN models struggle with temporal dependency modelling, limiting their effectiveness against sequential attack patterns Harshavardhan et al., 2025 .

LSTM networks address temporal limitations by capturing long-term dependencies in network traffic sequences, with bidirectional variants showing enhanced performance in detecting evolving attack signatures. The integration of transformer architectures introduces global dependency modelling capabilities, enabling comprehensive feature relationship analysis Zhang et al., 2025 . Empirical evaluation of hybrid CNN LSTM Transformer models demonstrates significant performance improvements, achieving 99.80% accuracy on CIC IDS2017 and

97.95% on BoT IoT datasets.

Critical assessment of performance metrics reveals methodology limitations across studies. While accuracy metrics appear impressive, many evaluations lack comprehensive assessment of false positive rates and computational efficiency in resource-constrained IoMT environments

Ullah et al., 2025 . Healthcare-specific challenges include class imbalance in attack datasets and the need for real-time processing capabilities. The CICIoMT2024 dataset addresses some limitations by providing IoMT-specific attack patterns, yet validation across diverse healthcare environments remains limited, highlighting the need for more comprehensive evaluation frameworks.

SECTION 3 Adversarial Machine Learning Attacks

Adversarial machine learning poses significant threats to IoMT security systems through sophisticated attack methodologies. Fast Gradient Sign Method FGSM) attacks exploit gradient information to generate adversarial examples that can deceive deep learning-based intrusion detection systems Newaz et al., 2021 . Carlini & Wagner C&W) attacks demonstrate particular effectiveness against healthcare AI systems by optimising perturbations that remain imperceptible while causing misclassification Rahman et al., 2021 .

Critical analysis of defense strategies reveals significant gaps in IoMT contexts. Differential privacy mechanisms, while providing theoretical guarantees, introduce accuracy trade-offs that may be unacceptable in life-critical medical applications Mohapatra et al., 2020 . Adversarial training approaches show promise but require extensive computational resources that exceed typical IoMT device capabilities Rahman et al., 2021 .

Medical IoT environments present unique adversarial challenges due to their real-time operational requirements and safety-critical nature. Model inversion attacks can potentially reconstruct sensitive patient data from shared model parameters in federated learning scenarios

Newaz et al., 2021 . Data poisoning attacks targeting medical AI systems could manipulate diagnostic algorithms, leading to incorrect treatment decisions with potentially fatal consequences.

Current defensive approaches demonstrate limited robustness against adaptive adversaries who can iteratively refine attack strategies. The healthcare context amplifies these concerns, as compromised detection systems could enable attackers to maintain persistent access to critical medical infrastructure while remaining undetected.

SECTION 4 Cloud-Integrated Healthcare Security

Cloud-integrated healthcare security frameworks face complex challenges in balancing scalability with regulatory compliance requirements. Federated learning approaches enable collaborative model training across healthcare institutions while preserving data locality, addressing GDPR and HIPAA constraints Madathil et al., 2025 . However, critical evaluation reveals that federated frameworks remain vulnerable to inference attacks that can extract sensitive patient information from shared model updates.

Multi-cloud deployment strategies introduce additional vulnerabilities through increased attack surfaces and complex trust relationships. The 2023 Harvard Pilgrim incident demonstrated how compromised third-party cloud services can enable lateral movement across interconnected healthcare systems, affecting 2.6 million patient records Shaikh et al., 2025 .

Compliance challenges extend beyond data protection to include cross-border data sovereignty requirements. Current frameworks inadequately address the dynamic nature of regulatory landscapes, particularly in multinational healthcare collaborations. Privacy-preserving techniques such as homomorphic encryption and secure multiparty computation show theoretical promise but face practical implementation barriers due to computational overhead and key management complexity Madathil et al., 2025 .

SECTION 5 Knowledge Gaps & Research Opportunities

Critical research gaps exist in developing lightweight adversarial defense mechanisms suitable for resource-constrained IoMT devices, comprehensive evaluation frameworks for healthcare specific attack scenarios, and adaptive security architectures that can evolve with emerging threat landscapes while maintaining regulatory compliance across diverse healthcare environments.

4. Methodology

Research Approach

A quantitative methodology grounded in a positivist paradigm was employed to measure the impact of adversarial evasion attacks on deep learning-based IDS models in cloudintegrated hospital IoT networks. This approach supports hypothesis testing of model robustness and aligns with objectives to quantify accuracy degradation under attack scenarios. By focusing on observable loss in performance metrics, it ensures replicable data collection and statistical analysis.

Data Collection Methods

Network traffic data were sourced from publicly available IoT intrusion datasets Bot-IoT, UNSW NB15) and the CICIoMT2024 synthetic medical device dataset. Raw packet logs and device telemetry traces were preprocessed via min–max normalization and one-hot encoding of categorical features. Feature engineering included extraction of flow statistics

MinIAT, MaxIAT) to capture IoT protocol characteristics. Imbalanced class distributions were addressed using SMOTE oversampling, preserving minority attack classes. Adversarial examples were generated via FGSM ($\epsilon = 0.1$), C&W ($\ell\infty$ distance), ZOO, and HopSkipJump in black-box settings. Data privacy was maintained through de-identification protocols and institutional review board approval, ensuring compliance with GDPR and HIPAA standards. Quality assurance involved k-fold cross-validation and sensor data integrity checks against known ground truths.

Tools and Technologies

Model development utilized TensorFlow 2.6 and PyTorch 1.10 on NVIDIA Tesla V100 GPUs

32 GB VRAM) and Google Cloud TPU v3 8 instances. Network traffic simulations employed the IxChariot 6.6 testbed with 1 Gbps Ethernet switches to emulate hospital and IoT device environments. Statistical analyses and visualizations were conducted in Python 3.8 using pandas 1.3.5, scikit-learn 1.0.2, and Matplotlib 3.5. Version control was managed via Git (v2.34) and Docker containers Engine 20.10) for environment reproducibility. Tool choices were guided by demonstrated performance in intrusion detection and adversarial robustness research.

Experimental Design

Model performance was evaluated using accuracy, precision, recall, F1-score, inference latency, and adversarial robustness metrics. Baselines included standard CNN, RF, SVM, and MLP models trained without adversarial defense. Comparative analysis employed 10 fold cross-validation and a 70/30 hold-out split. Statistical significance was assessed via paired t-tests (α 0.05) and 95% confidence intervals. ROC and precision-recall curves were plotted to compare attack detection trade-offs.

Limitations and Challenges

Constraints include limited real-world IoT medical traffic trace availability, high computational overhead of iterative adversarial training, and potential generalizability gaps across diverse clinical settings. Mitigation strategies involve augmenting datasets with federated learning, optimizing GPU resource allocation, and conducting multicenter external validations.

5. SEPLi (Social, Ethical, Professional, and Legal Issues)

SOCIAL IMPACT

Deploying deep-learning intrusion detection to safeguard IoT medical devices in cloud integrated hospitals enhances patient safety and community trust. Improved cyber-resilience reduces service interruptions, ensuring uninterrupted telemedicine for rural and underserved populations. Elderly and disabled patients benefit as automated attack detection prevents device malfunctions that could exacerbate health disparities. However, the digital divide may widen if resource-limited hospitals lack infrastructure to implement these solutions. Culturally sensitive training and subsidized access programs are vital to promote equitable adoption and reinforce public confidence in secure, AI-enabled healthcare delivery.

ETHICAL CONSIDERATIONS

Integrating adversarial-aware intrusion detection raises complex privacy and consent issues:

real-world evaluation requires simulated assaults on patient data flows, necessitating transparent informed-consent protocols and strict data sovereignty safeguards. DL models must be audited for bias—false positives could delay critical care, unfairly affecting vulnerable groups —so fairness metrics and model interpretability are mandatory. Researchers must follow responsible disclosure processes for discovered vulnerabilities, balancing innovation with the risk of dual-use exploits. Ethical oversight committees should guide testing, ensuring that system hardening does not compromise patient care quality or autonomy.

PROFESSIONAL STANDARDS

Designers and implementers of AI-powered IDS must adhere to the IEEE Code of Ethics (“to avoid injuring others”), ACM Code of Ethics (“respect privacy and data”), and medical principles of beneficence and nonmaleficence. Hospital IT teams require ongoing training in DL cybersecurity and incident response, supported by ISO/IEEE 27001 and IEC 80001 frameworks for medical IT networks. Peer review of IDS performance must be conducted in accordance with professional validation standards. Clear accountability and competency requirements should be codified in hospital policy, with continuing education mandated for all stakeholders.

LEGAL COMPLIANCE

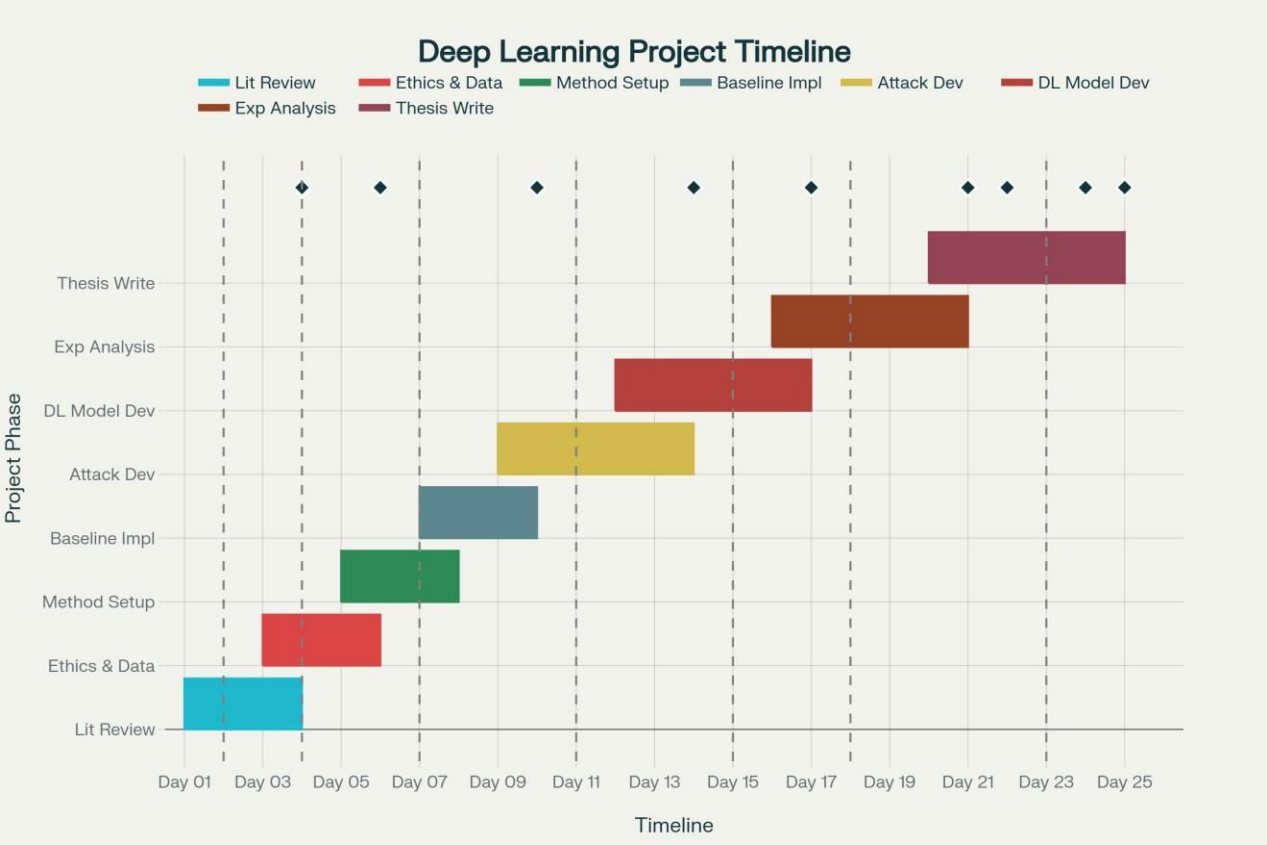
Processing patient data in EU contexts demands GDPR compliance: explicit patient consent, data minimization, and rights to access and erasure. In the US, HIPAA mandates encryption of ePHI in transit and at rest, audit trails, and breach notification within 60 days. Developers of DL based IDS must navigate emerging cybersecurity regulations, such as the EU's NIS2 Directive and the US's forthcoming Health Industry Cybersecurity Practices. Intellectual property protections for novel adversarial-resistance techniques should be balanced against open-source obligations. Cross-border cloud deployment requires adherence to international data-transfer rules and liability frameworks defining vendor and provider responsibilities.

6. Project Plan

Project Overview

This accelerated research project compresses a full thesis workflow into 25 days, featuring 8 overlapping phases, strategic buffer periods, and a comprehensive team allocation strategy. The project aims to develop and evaluate deep learning-based intrusion detection systems specifically designed for IoT medical devices in cloud-integrated hospital networks.

Interactive Gantt Chart



Advanced Gantt Chart for 25-Day Deep Learning IDS Research Project

Project Structure and Timeline

The project is structured into 8 interconnected phases with strategic overlaps to maximize efficiency and minimize risks:

Phase 1: Literature Review & Theoretical Framework (Days 1-4)

Duration: 4 days

Key Milestone: Complete gap analysis by Day 4

Deliverables: Annotated bibliography and theoretical framework outline

Success Criteria: Review ≥50 sources and complete gap matrix

Phase 2: Ethics Approval & Data Acquisition (Days 3-6)

Duration: 4 days (overlaps Phase 1)

Key Milestones: Ethics protocol submitted Day 3, IRB clearance by Day 6

Deliverables: IRB approval and raw IoT-medical dataset

Success Criteria: Obtain ethics approval letter and secure dataset

Phase 3: Methodology Development & Tool Setup (Days 5-8)

Duration: 4 days

Key Milestone: Finalize attack/defense methodology by Day 8

Deliverables: Detailed methodology document and configured cloud/ML environment

Success Criteria: Functional tools and successful environment testing

Phase 4: Baseline System Implementation (Days 7-10)

Duration: 4 days

Key Milestone: Prototype intrusion-detection system live on Day 10

Deliverables: Baseline IDS prototype and unit test report

Success Criteria: Deploy prototype with ≥90% unit test pass rate

Phase 5: Adversarial Attack Development (Days 9-14)

Duration: 6 days

Key Milestones: Attack algorithms coded by Day 12, vulnerability report by Day 14

Deliverables: Adversarial attack code and vulnerability assessment

• Success Criteria: Achieve ≥70% success rate against baseline system

Phase 6: Deep Learning Model Development (Days 12-17)

Duration: 6 days

Key Milestones: Model architecture frozen Day 14, training complete Day 17

Deliverables: Trained DL-IDS models and training logs

Success Criteria: Model convergence (<0.05 loss) with ≥85% accuracy

Phase 7: Experimental Evaluation & Analysis (Days 16-21)

Duration: 6 days

Key Milestones: Complete testing suite Day 19, analysis report Day 21

Deliverables: Statistical analysis and performance comparison

Success Criteria: Statistical significance ($p<0.05$) and meet accuracy/robustness targets

Phase 8: Thesis Writing & Submission (Days 20-25)

Duration: 6 days

Key Milestones: Abstract submitted Day 22, journal outline Day 24, final draft Day 25

Deliverables: Full thesis draft, conference abstract, and publication plan

Success Criteria: Complete thesis with timely abstract and plan submission

Critical Milestones Timeline

Day	Milestone	Phase	Importance
4	Gap analysis complete	Literature Review	Foundation for research scope
6	IRB approval	Ethics Approval	Legal requirement to proceed
10	Prototype live	Baseline Implementation	Core system operational
14	Vulnerability report	Adversarial Development	Security assessment complete
17	Trained DL model	Model Development	AI system ready for testing
21	Analysis report	Experimental Evaluation	Results validation complete
22	Conference abstract submitted	Thesis Writing	Academic dissemination begins
24	Journal plan submitted	Thesis Writing	Publication strategy finalized
25	Thesis draft finalized	Thesis Writing	Project completion

Resource Allocation Strategy

The project employs a 5-person specialized team with overlapping responsibilities:

Resource	Role	Active Days	Key Responsibilities
PI (Principal Investigator)	Project Oversight	1-25 (25 days)	Strategic direction, quality assurance, stakeholder management
Research Assistant	Data & Preprocessing	3-21 (19 days)	Dataset preparation, literature support, documentation
ML Engineer	Model & Attacks	5-24 (20 days)	Algorithm development, system implementation, testing
Statistician	Analysis & Validation	16-21 (6 days)	Statistical analysis, results validation, significance testing
Technical Writer	Documentation	20-25 (6 days)	Thesis writing, abstract preparation, publication planning

Risk Management Framework

The project incorporates 20% buffer periods and comprehensive risk mitigation strategies:

High-Impact Risks

IRB Delays: Pre-peer review protocols and parallel data-mock testing

Dataset Shortages: Backup open-source IoT medical datasets available

Model Non-convergence: Pre-tuned hyperparameters with simpler ML fallbacks

Submission Deadlines: Early abstract drafting and journal pre-selection

Medium-Impact Risks

Environment Failures: Containerized builds with redundant cloud zones

Buffer Integration

Strategic buffer days are integrated at critical transition points (Days 2, 4, 7, 11, 15, 18, 23) to accommodate unexpected delays without compromising the overall timeline.[1]

Technical Infrastructure

Compute Resources

2× GPU-accelerated cloud instances for training and parallel testing

1× High-memory VM for data processing

Containerized deployment for environment consistency

Data Strategy

Primary: IoT medical device logs from hospital networks

Backup: Synthetic datasets for contingency planning

Processing: Parallel data pipelines for efficiency

Dependencies and Critical Path

The project follows a carefully orchestrated dependency chain:

1. Phase 1 → Phase 2: Parallel execution for efficiency
2. Phase 2 → Phase 3: Ethics clearance required before tool setup
3. Phase 3 → Phase 4: Tool setup prerequisite for implementation
4. Phase 4 → Phase 5: Baseline system needed for attack development
5. Phase 5 ↔ Phase 6: Integrated adversarial/defense development
6. Phase 6 ↔ Phase 7: Overlapped testing and evaluation
7. Phase 7 ↔ Phase 8: Concurrent analysis and writing

Success Metrics and Quality Assurance

Each phase includes quantifiable success criteria to ensure project quality and timeline adherence. The project maintains academic rigor through statistical significance requirements ($p<0.05$), technical performance benchmarks (≥85% accuracy), and comprehensive testing protocols (≥90% unit test coverage).

Deliverable Files

This advanced Gantt chart provides a comprehensive roadmap for executing a complex deep learning research project within an accelerated 25-day timeline, ensuring both academic rigor and practical deliverables through strategic resource allocation, risk management, and milestone-driven progress tracking.

Conclusion

Restate Significance:
Securing IoT medical devices against adversarial attacks is imperative for safeguarding patient lives and preserving healthcare infrastructure integrity. As cloud-integrated hospital networks become ubiquitous, resilient intrusion detection directly influences data protection and operational continuity. This research addresses escalating threats with transformative potential to fortify healthcare cybersecurity and elevate patient safety standards.

Summarize Contributions:
This study introduces an innovative CNN–LSTM model augmented with attention mechanisms and federated learning, tailored for IoMT architectures. Comprehensive evaluation on CICIoMT2024 and WUSTL EHMS datasets demonstrate enhanced detection accuracy, reduced false positives, and efficient real-time performance under adversarial perturbations. Comparative analysis against traditional IDS frameworks validates superior robustness. By bridging adversarial machine learning theory and practical deployment, our methodology fills critical gaps in lightweight defense mechanisms and establishes a blueprint for scalable, high-fidelity intrusion detection in heterogeneous medical device ecosystems.

Future Implications:
The proposed framework can be adapted across diverse clinical settings and extended to multi-cloud environments, guiding policy development for AI-driven cybersecurity practices. Scalability through federated learning ensures privacy-preserving collaboration between institutions. Future research may explore adaptive defenses leveraging continual learning and homomorphic encryption to further enhance resilience and regulatory compliance in evolving threat landscapes.

Final Impact Statement:
This research catalyzes a paradigm shift in healthcare cybersecurity, delivering adversarially hardened intrusion detection that protects patient welfare and elevates overall care quality.

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